

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 07/01/2026**
Time: 3 Hrs **GTM - 11** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A travelling wave is represented by equation $\psi\left(\vec{r},t\right)=5\sin\left(7t-x+2y-\sqrt{20}z\right)$

The angle between direction of propagation of wave and positive x-axis

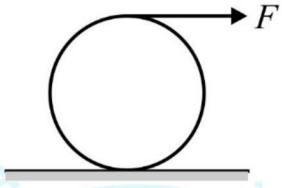
A) $\cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$ B) $\cos^{-1}\left(\frac{1}{5}\right)$ C) $\cos^{-1}\left(\frac{2}{5}\right)$ D) $\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$

2. The dimensions of $\frac{a}{b}$ in the equation $P=\frac{a-t^2}{bx}$, Where P is pressure, x is distance and t is time, are

A) $\left[M^2LT^{-3}\right]$ B) $\left[MT^{-2}\right]$ C) $\left[LT^{-3}\right]$ D) $\left[ML^3T^{-1}\right]$

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3. A force F is applied at the top of a ring of mass M and radius R placed on a rough horizontal surface as shown in figure. Friction is sufficient to prevent slipping. The friction force acting on the ring is



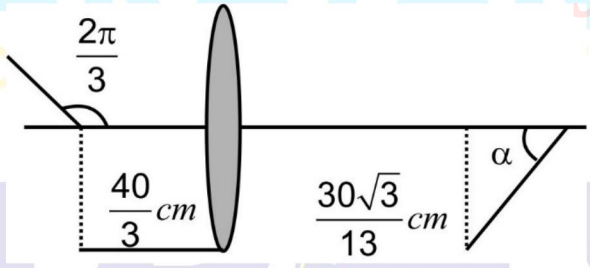
A) $\frac{F}{2}$ towards right

B) $\frac{F}{2}$ towards left

C) $\frac{2F}{3}$ towards right

D) ZERO

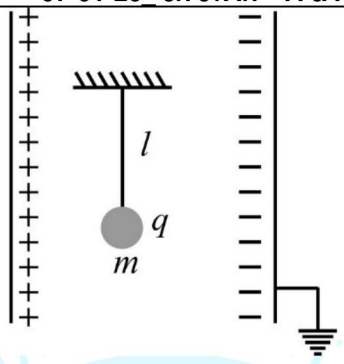
4. A rod of length 2 cm makes an angle $\left(\frac{2\pi}{3}\right) rad$ with the principal axis of a thin convex lens. The lens has a focal length of 10 cm and is placed at a distance of $\frac{40}{3} cm$ from the object as shown in the figure. The height of the image is $\frac{30\sqrt{3}}{13} cm$. The angle made by image with respect to the principal axis is



A) $\pi/2$ B) $\pi/3$ C) $\pi/4$ D) $\pi/6$

5. A simple pendulum has a length l, mass of bob m. The bob is given a charge q. The pendulum is suspended between the vertical plates of charged parallel plate capacitor. If E is the field strength between the plates, then time period T for small oscillations equals

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A) $2\pi\sqrt{\frac{l}{g}}$

B) $2\pi\sqrt{\frac{l}{g+\frac{qE}{m}}}$

C) $2\pi\sqrt{\frac{l}{g-\frac{qE}{m}}}$

D) $2\pi\sqrt{\frac{l}{\sqrt{g^2+\left(\frac{qE}{m}\right)^2}}}$

6. A man wishes to swim across a river 400m wide flowing with a speed of 3m/s, such that he reaches the point just infront on the other bank in time not greater than 100s. The angle made by the direction he swims and river flow direction is

A) 53^0 B) 127^0 C) 90^0 D) 37^0

7. Consider a point charge +q placed at the origin and another point charge –2q is placed at a distance of 9 m from the charge +q along the x-axis. Determine the point between the two charges at which electric potential is zero.

A) 6m from +q B) 3.727m from +q

C) 3m from +q D) 5.272m from +q

8. A power bank, generally used to charge a mobile phone, consists of battery capable of supplying 3A current at 5V for 2 hours. If efficiency of powerbank is 85%, then amount of energy wasted in full discharge of power bank is

A) 4.5 J B) 4.5 KJ C) 16.2 J D) 16.2 KJ

9. Vertical displacement of a plank with a body of mass m on it is varying according to law $y=\sin(\omega t)+\sqrt{3}\cos(\omega t)$. The minimum value of ω for which the mass just breaks off the plank and the moment it occurs first after $t = 0$, are given by Assume y to be positive vertically upwards.

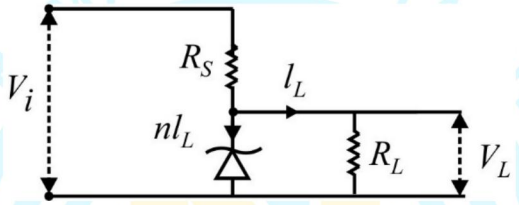
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A) $\sqrt{\frac{g}{2}}, \frac{\pi}{6}\sqrt{\frac{2}{g}}$ B) $\frac{g}{\sqrt{2}}, \frac{2}{3}\sqrt{\frac{\pi}{g}}$ C) $\sqrt{\frac{g}{2}}, \frac{\pi}{3}\sqrt{\frac{2}{g}}$ D) $\sqrt{2g}, \sqrt{\frac{2\pi}{3g}}$

10. Relative permeability of iron is 5500, then its magnetic susceptibility will be

A) 5500×10^{7} B) 5500×10^{-7} C) 5501 D) 5499

11. The value of the resistor, R_s, needed in the dc voltage regulator circuit shown here, equals



A) $\frac{(V_i - V_L)}{nI_L}$ B) $\frac{(V_i + V_L)}{nI_L}$ C) $\frac{(V_i - V_L)}{(n+1)I_L}$ D) $\frac{(V_i + V_L)}{(n+1)I_L}$

12. Two gases, initially having the same pressure, volume and temperature expand to the same final volume, the first isothermally and the second adiabatically. Which of the following is true?

A) Greater work is done by the gas for isothermal process.

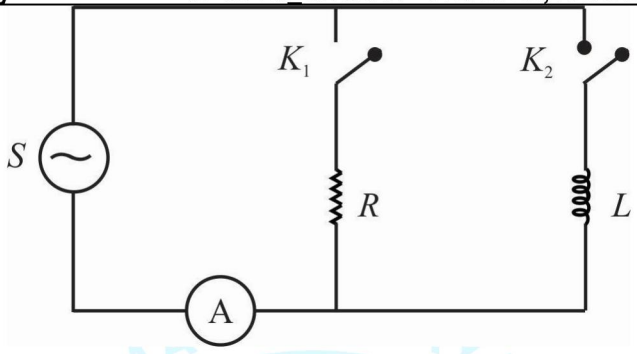
B) The isothermal process has lower final pressure.

C) The isothermal process has lower final temperature

D) Data is insufficient to arrive at a conclusion.

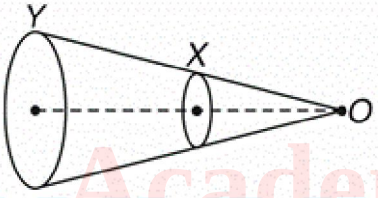
13. In the given circuit, R is a pure resistor, L is a pure inductor. S is a 100V, 50HZ ac source and A is an ac ammeter. With either K_1 or K_2 alone closed, the ammeter reading is I. If the source is changed to 100V, 100HZ, the ammeter reading when K_1 alone is closed and with K_2 alone is closed will be respectively,

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A) $I, \frac{I}{2}$ B) $I, 2I$ C) $3I, I$ D) $2I, \frac{I}{2}$

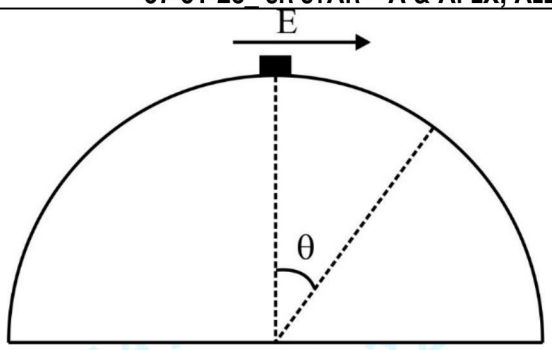
14. Two circular coils X and Y have equal number of turn and carry equal currents in the same sense and subtend same solid angle at point O. If the smaller coil X is midway between O and Y, then if we represent the magnetic induction due to bigger coil Y at O as B_Y and that due to smaller coil X at O as B_X, then



A) $\frac{B_Y}{B_X} = 1$ B) $\frac{B_Y}{B_X} = 2$ C) $\frac{B_Y}{B_X} = \frac{1}{2}$ D) $\frac{B_Y}{B_X} = \frac{1}{4}$

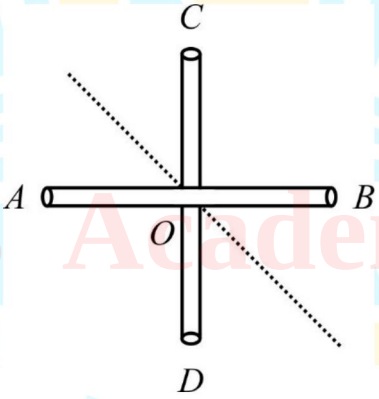
15. A positively charged small disc of mass m is released from the top of fixed hemispherical frictionless dome in presence of a uniform horizontal electric field of magnitude E as shown in figure. If the disc leaves the dome after an angular displacement $\theta = sin^{-1}\left(\frac{3}{5}\right)$, find ratio of gravitational and electrostatic forces on the disc. Assume that the dome does not exhibit any electrical property.

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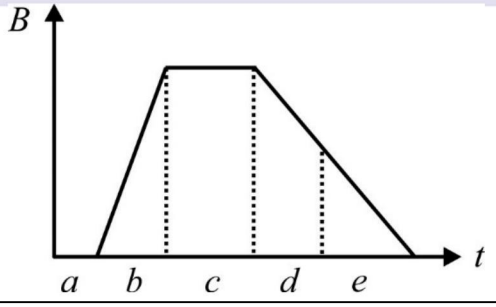
A) $\frac{4}{3}$ B) $\frac{3}{4}$ C) $\frac{9}{2}$ D) 12

16. *AB* and *CD* are two identical rods each of length *l* and mass *m* joined to form a cross. The moment of inertia of these two rods about a bisector of the angle between the rods is



A) $\frac{ml^2}{6}$ B) $\frac{ml^2}{3}$ C) $\frac{ml^2}{12}$ D) $\frac{2ml^2}{3}$

17. The graph gives the magnitude $B(t)$ of a uniform magnetic field that exists throughout a conducting loop, perpendicular to the plane of the loop. Rank the five regions of the graph according to the magnitude of the emf induced in the loop, greatest first



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A) $b > (d = e) < (a = c)$ B) $b > (d = e) > (a = c)$

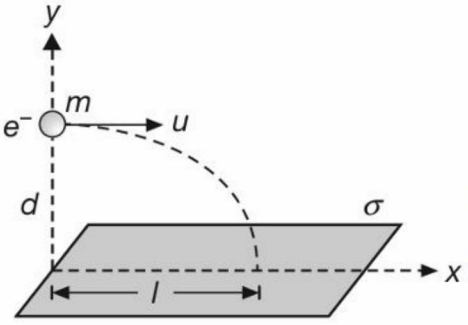
C) $b < d < e < c < a$ D) $b > (a = c) > (d = e)$

18. Match List I with List II and select the correct answer using the codes given below the lists

List I	**List II**
P. Boltzmann constant	1. $[ML^2T^{-1}]$
Q. Coefficient of viscosity	2. $[ML^{-1}T^{-1}]$
R. Planck constant	3. $[MLT^{-3}K^{-1}]$
S. Thermal conductivity	4. $[ML^2T^{-2}K^{-1}]$

Codes :	P	Q	R	S
A)	3	1	2	4
B)	3	2	1	4
C)	4	2	1	3
D)	4	1	2	3

19. An electron is projected near surface at a distance d and with initial velocity u parallel to a uniformly charged flat conducting plate as shown. It strikes the plate after travelling a distance l along the direction of projection. The surface charge density σ of the conducting plate is (Ignore gravity)



A) $\sigma = \dfrac{d \in_0 mu^2}{el}$ B) $\sigma = \dfrac{d \in_0 mu}{el}$

C) $\sigma = \dfrac{2d \in_0 mu^2}{el^2}$ D) $\sigma = \dfrac{2d \in_0 mu}{el}$

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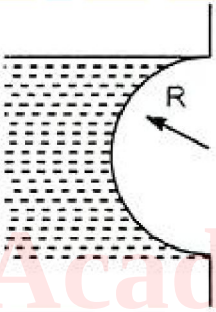
20. The kinetic energy of electrons is E, when the incident light has wavelength λ. To increase the K.E. to $2E$, the incident light must have wavelength

A) $\frac{hc}{E\lambda - hc}$ B) $\frac{hc\lambda}{E\lambda + hc}$ C) $\frac{h\lambda}{E\lambda + hc}$ D) $\frac{hc\lambda}{E\lambda - hc}$

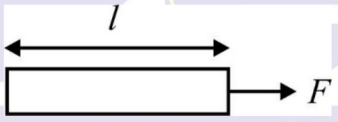
SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. A hemisphere of radius R is just submerged in water of density 'ρ'. The total hydrostatic force acting on hemisphere is $\frac{\sqrt{x}}{y}\rho\pi R^3 g$, then find $x+y$



22. A uniform rod of length l is acted upon by a force F in a gravity-free region, as shown in the figure. If the area of cross-section of the rod is A and it's Young's modulus is Y, then the elastic potential energy stored in the rod due to elongation is $\frac{F^2 l}{nA^2 Y}$, then the value of n is____



23. First diffraction minimum due to a single slit of width 10^{-4} cm is at $\theta = 30^0$. The wavelength of the light used is (in n m)

24. In a sample of gas containing non-linear polyatomic molecules having 3 vibrational modes, the ratio of vibrational energy to translation kinetic energy is equal to

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25. If a body travels in a straight line, covering the first half of total distance with a speed 3m/s and second half of distance with a speed 6m/s, then the average speed over the entire journey is_____ (in m/s),

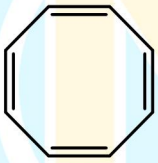
CHEMISTRY MAX.MARKS: 100

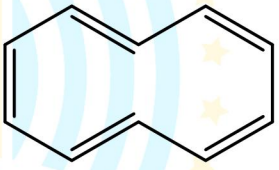
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

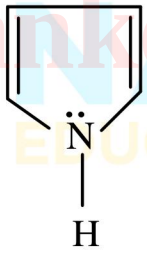
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Which of the following is correct regarding the following compounds / ions

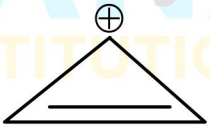

(a)

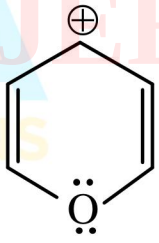

(b)


(c)


(d)


(e)


(f)


(g)

A) a,d,e,f,g Aromatic and b,c Non-aromatic

B) a,d,e,f,g Non-aromatic and b,c Aromatic

C) a,d,e –Aromatic and b,c,f,g – Non-aromatic

D) b,c,f – Aromatic and a,d,e – Non -aromatic

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27. An optically active alkyl halide C_4H_9Cl (A) reacts with tertiary butoxide to give alkene (B) as the major product (B) react with HBr in presence of peroxide to form (C) by anti-markovnikov addition (C) react with aqueous KOH to form Alcohol (D) oxidation of (D) acidic $KMnO_4$ gives compound (E) The IUPAC name of compound (E) is

A) Propanoic acid B) Butanoic acid

C) Ethanoic acid D) 2-Mehtyl propanoic acid

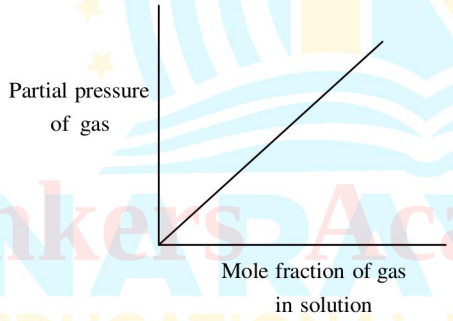
28. Which of the following property of 17 group (halogens) generally increases down the group

A) Electronegativity B) First Ionisation energy

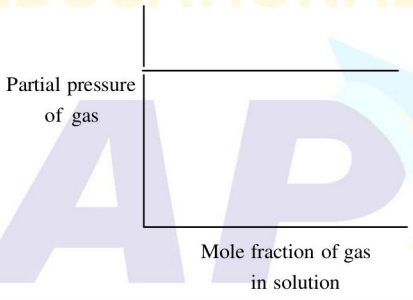
C) Boiling point D) Electron affinity

29. Which of the following curves represent the Henry's law

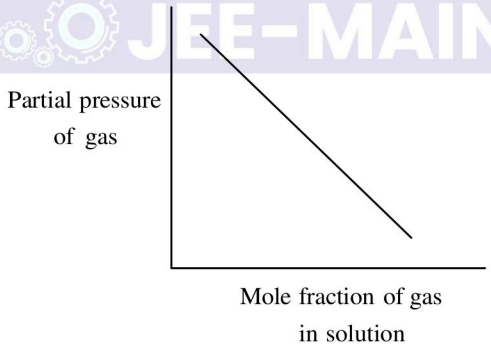
A)



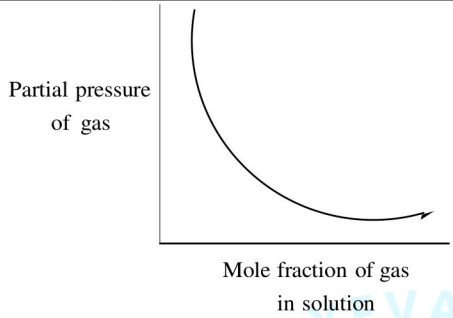
B)



C)



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D)

30. According to Bohr's model of hydrogen atom which of the following statements is incorrect

A) Radius of 5th orbit is 25 times larger than that of 1st orbit

B) Radius of 6th orbit is 9 times larger than that of 2nd orbit

C) Radius of 4th orbit is 4 times larger than that of 2nd orbit

D) Radius of 2nd orbit is 2 times larger than that of 1st orbit

31. If water is assumed to be a perfect gas, molar enthalpy change for vapourisation of 1mole of water at 1 bar and $100^0 C$ is $41 KJ\, mole^{-1}$. The internal energy change when 1mole of water is vapourised at 1 bar pressure and $100^0 C$ nearly in $KJ / mole$ is $(R = 8.3)$

A) 41.0 B) 44.0 C) 86.0 D) 38.0

32. A solution is made by mixing one mole of volatile liquid A with 3 moles of volatile liquid "B" the vapour pressure of pure "A" is 100mm Hg and that of the solution is 500mm Hg. The vapour pressure of pure B and the least volatile component of the solution respectively

A) 1400 mm Hg, A B) 1400 mm Hg, B

C) 633 Hg, B D) 633 Hg , A

33. $CaCO_3(s)+2HCl(aq)\rightarrow CaCl_2(aq)+CO_2(g)+H_2O(l)$ If 500 gr of $CaCO_3$ react with 300 ml of 1.20 M HCl what mass of $CaCl_2$ will be formed (Given molar masses (Ca = 40, C = 12, O = 16, H = 2, Cl = 35.5 gr/mol)

A) 19.98g B) 111g C) 27.75g D) 9.99g

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34. The solubility product of sparingly soluble salt A_2B_3 type is 3.456×10^{-12}. The solubility is (in moles/lit)

A) 1.3×10^{-6} B) 2×10^{-5} C) 2×10^{-3} D) 4.5×10^{-4}

35. The hybridisation of the molecule having a non-zero dipole moment and the maximum number of lone pairs on the central atom Among the following molecules

ClO_2, PCl_3, BrF_3, SF_6, XeF_2 is_____

A) sp^3 B) sp^3d C) sp^2 D) sp^3d^2

36. Statement I : Salicylaldehyde reacts with NaOH but does not give Tollen's test

Statement II : Salicylaldehyde does not undergo self – aldol condensation

A) I incorrect, II correct B) I correct, II incorrect

C) Both are Incorrect D) Both are Correct

37. Choose the correct statement

A) Tryptophan and tyrosine are aromatic amino acids

B) Leucine is the only amino acid with two chiral centres

C) Aspartic acid contains an –OH group

D) Valine contains a sulphur atom

38. Arrange the following in increasing order of basic nature in (aqueous)

NH_3, CH_3NH_2, $(CH_3)_2NH$, $(CH_3)_3N$

A) $NH_3<(CH_3)_3N<CH_3NH_2<(CH_3)_2NH$

B) $NH_3<CH_3NH_2<(CH_3)_3N<(CH_3)_2NH$

C) $CH_3NH_2<NH_3<(CH_3)_3N<(CH_3)_2NH$

D) $(CH_3)_3N<CH_3NH_2<NH_3<(CH_3)_2NH$

39. Statement I: The metallic radius of In is greater than that of Ga

Statement II: The ionic radius of Ga^{+3} is greater than that of Al^{+3}

Choose the correct answer

A) Both statement I and Statement II are incorrect

B) Statement I is incorrect but Statement II is correct

C) Statement I is correct but Statement II is incorrect

D) Both Statement I and Statement II are correct

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40. Statement I: In octahedral complexes, strong field ligands like $CN^{\ominus}$ and CO produce low spin complexes because they cause a large splitting (Δ_o)

Statement II: In tetrahedral complexes Δ_t is always smaller than pairing energy (P) hence tetrahedral complexes are almost always high spin.

Choose the correct option

A) Statement I is correct but Statement II is incorrect

B) Statement I is incorrect but Statement II is correct

C) Both Statement I and Statement II are incorrect

D) Both Statement I and Statement II are correct

41. Choose the correct tests with respective observations

(1) $FeCl_3$ solution $+K_4\left[Fe(CN)_6\right] \rightarrow$ Blue precipitate

(2) $CoCl_2$ solution $+KNO_2 + CH_3COOH \rightarrow$ Yellow precipitate

(3) $CuSO_4$ solution $+CH_3COOH + K_4\left[Fe(CN)_6\right] \rightarrow$ Blood red colour

(4) $BaCO_3$ solution $+CH_3COOH + K_2CrO_4 \rightarrow$ yellow precipitate

(5) $PbCl_2$ (hot solution) $+K_2CrO_4 \rightarrow$ Black precipitate

A) 1, 2 and 5 only B) 2, 3 and 4 only

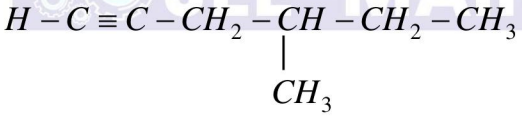
C) 1, 2 and 4 only D) 3, 4 and 5 only

42. On complete combustion, 1.50 g of an organic compound (X) produced 2.20 g of CO_2 and 0.900 g of H_2O. The empirical formula mass of the compound (X) is

(Given atomic masses: C = 12, H = 1, O = 16)

A) 30 B) 44 C) 60 D) 29

43. IUPAC name of the following compound is___

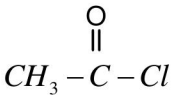


A) 3 – Methylhex – 6 – yne B) 4 – Methylhex – 1 – yne

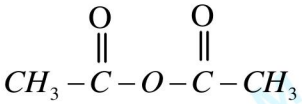
C) 4 – Methylhex – 2 – yne D) 3 – Methylhept – 5 – yne

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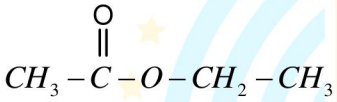
44. Consider the following molecules



(p)



(q)



(r)



(s)

The correct order of rate of hydrolysis

A) r > p > q > s

B) p > q > r > s

C) q > p > s > r

D) p > r > q > s

45. A molecule has the formula AX_4Y_2 and all atoms belong to p-block. Element A is a rare, monoatomic, non-radioactive noble gas of its group. It has the lowest ionization enthalpy among A, X, and Y.

Element X and Y have the highest and second-highest electronegativity values among all elements

The shape of the molecule is

A) Square planar

B) Square pyramidal

C) Octahedral

D) T-shaped

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

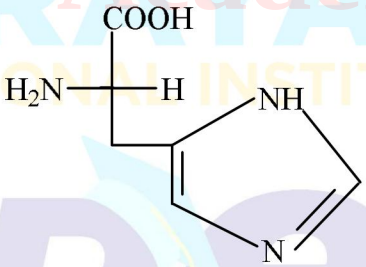
46. A transition metal (M) among V, Cr, Mn, and Fe has the highest standard electrode potential (M^{3+}/M^{2+})

It forms a metal complex of the type $\left[M(CN)_6\right]^{3-}$ The number of electrons present in the e_g orbital of the complex is

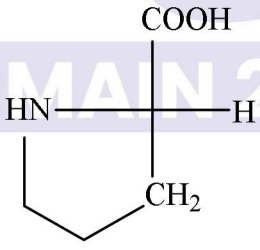
47. The following reaction takes place in a cell $Mg_{(s)} + 2Ag^{+}(0.0001M) \rightarrow Mg^{+2}(0.10M) + 2Ag_{(s)}$ the value of E_{cell} of the following cell $Mg\left|Mg^{+2}(0.10M)\right|\left|Ag^{+}(0.0001M)\right|Ag$ is $x \times 10^{-2}V$, then the value of $'x'$ is____

(given $E^0_{cell} = 3.17V$)

48. (i) The weight of 1mole of the following Amino Acid is $'x'\,gr$



(ii) The weight of 1mole of the following compound called proline is $'y'\,gr$



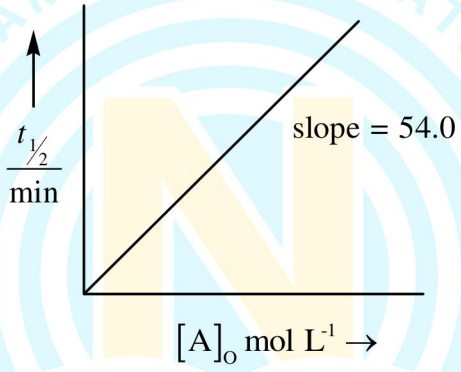
Therefore $\frac{x+y}{10} = z$. The value of $'z'$ is___

(Given : molar mass in g mol-1 H : 1, C : 12, N : 14, O : 16)

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49. The partial pressure of $CH_3OH(g), CO(g)$ and $H_2(g)$ in equilibrium mixture for the reaction, $CO(g)+2H_2(g) \rightleftharpoons CH_3OH(g)$ are 2.0, 1.0 and 0.1 atm respectively at 427^oC. The value of K_p for the decomposition of CH_3OH into CO and H_2 is _____ $\times 10^{-3} atm^2$

50. For the reaction A $\rightarrow$ products



A graph of $t_{1/2}$ (minutes) vs $[A]_0 (mol\ L^{-1})$ is a straight line with a slope = 54.0

If the reaction was started with $[A]_0 = 3.0\ mol\ L^{-1}$ then $t_{\frac{1}{2}}$ of the reaction is (in minutes)

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Number of zeros in the expansion of 100! is

A) 22 B) 23 C) 24 D) 25

52. The ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the hyperbola $\frac{x^2}{A^2}-\frac{y^2}{B^2}=1$ are given to be confocal and length of minor axis of the ellipse is same as the conjugate axis of the hyperbola. If e_1 and e_2 represents the eccentricities of ellipse and hyperbola respectively, then the value of $e_1^{-2}+e_2^{-2}$ is

A) 1 B) 2 C) 0 D) $\frac{1}{2}$

53. The number of natural numbers between 212 to 999 such that sum of their digit is 15 is equal to

A) 63 B) 65 C) 62 D) 64

54. If $f\left(\frac{x+2y}{3}\right)=\frac{f(x)+2f(y)}{3}\forall x,y\in R$ and $f'(0)=1, f(0)=2$, then find $f(x)$.

A) $x+2$ B) $2x+1$ C) $x-2$ D) $2x-5$

55. Let $\alpha\in(0,\infty)$ and $A=\begin{bmatrix}1&2&\alpha\\1&0&1\\0&1&2\end{bmatrix}$. If $\det\left(adj\left(2A-A^T\right)\right).Adj\left(A-2A^T\right)=2^8$, then $\left(\det(A)\right)^2$ is equal to:

A) 49 B) 16 C) 36 D) 1

56. The term independent of x in the expansion of $\left(1-x^2\right)\left(x+\frac{2}{x}\right)^6$

A) 160 B) 240

C) –80 D) No constant terms exist

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57. If $2\cos\theta+2\sqrt{2}=3\sec\theta$ where $\theta\in(0,2\pi)$ then which of the following can be correct?

A) $\cos\theta=-\frac{1}{\sqrt{2}}$ B) $\tan\theta=-\sqrt{3}$ C) $\sin\theta=\frac{\sqrt{3}}{2}$ D) $\cot\theta=-1$

58. If $(1+x)^{10}=a_0+a_1x+a_2x^2+a_3x^3+....+a_{10}x^{10}$, then $(a_0-a_2+a_4-a_6+a_8-a_{10})^2+(a_1-a_3+a_5-a_7+a_9)^2$ is equal to

A) 3^{10} B) 2^{10} C) 2^9 D) 3^9

59. A polynomial function $p(x)=2x^3-3ax^2-12a^2x-2$ attains its local maxima at x = s and local minima at x = t. If $p\left(\frac{s+t}{2}\right)=-54$, then the value of 's' is

A) 2 B) –4 C) –2 D) 4

60. If the locus of $Z\in C$, such that $\text{Re}\left(\frac{z-1}{2z+i}\right)+\text{Re}\left(\frac{\bar{z}-1}{2\bar{z}-i}\right)=2$ is a circle of radius r and center (a,b) then $\frac{15ab}{r^2}$ is equal to :

A) 24 B) 12 C) 18 D) 16

61. Let $\vec{a}=\hat{i}+2\hat{j}+\hat{k}$, $\vec{b}=3\hat{i}-3\hat{j}+3\hat{k}$, $\vec{c}=2\hat{i}-\hat{j}+2\hat{k}$ and $\vec{d}$ be a vector such that $\vec{b}\times\vec{d}=\vec{c}\times\vec{d}$ and $\vec{a}.\vec{d}=4$. Then $\left|(\vec{a}\times\vec{d})\right|^2$ is equal to _____.

A) 128 B) 182 C) 281 D) 218

62. If Z is the set of integers. Then, the relation $R=\{(a,b):1+ab>0\}$ on Z is

A) Reflexive and transitive but not symmetric

B) Symmetric and transitive but not reflexive

C) Reflexive and symmetric but not transitive

D) An equivalence relation

63. If $\lim\limits_{x\to0}\frac{(\gamma-1)e^{x^2}+x^2\sin(\alpha x)}{\sin(2x)-\beta x}=3$ then $\alpha+2\beta+\gamma$ is equal to

A) 0 B) 1 C) 3 D) 5

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64. If the system of linear equations

$x-2y+z=-4$

$2x+\alpha y+3z=5$

$3x-y+\beta z=3$

has infinitely many solutions, then $12\alpha+13\beta$ is equal to

A) 60 B) 64 C) 54 D) 58

65. $P_n=\alpha^n+\beta^n, \alpha+\beta=1, \alpha.\beta=-1, P_{n-1}=11, P_{n+1}=29$, then $P_n^2=$

A) –187 B) 184 C) 324 D) 364

66. P is any point lying on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b)$ whose foci are S and S'.

If $\angle PSS' = \alpha$ and $\angle PS'S = \beta$, then the value of $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}$ is

A) $\frac{1+e}{1-e}$ B) $\frac{1+e^2}{1-e^2}$ C) $\frac{1-e}{1+e}$ D) 1

67. Let the foot of perpendicular from P (5,1,-3) on the line $L_1: x-1=y-2=z$ and $L_2: x-2=y=z-1$ is Q and R, respectively. The area of triangle PQR is equal to

A) $\frac{7}{2}$ B) $\frac{7}{\sqrt{2}}$ C) $\frac{7\sqrt{3}}{2}$ D) 7

68. Let $a_i; i=1,2,3,.....n$ denote integers in the domain of the function $f(x)=\sqrt{\log_{\frac{1}{2}}\left(\frac{4x-25}{x-21}\right)}$ where $a_i<a_{i+1}, i=1,2,3....(n-1)$. The line $L:\frac{2x-a_1}{4}=\frac{y+a_1}{a_2}=\frac{z-a_3}{a_5}$ meets the xy, yz and zx planes at A, B and C, respectively C^1 is the image of C in X-axis. The volume of the tetrahedron $OABC^1$ (0 is origin) is V then $90V =$ ____

A) 280 B) 290 C) 300 D) 310

69. If the order of matrix A is 3 and $|A|=3$ then the value of $\det\left(adj\left(-4adj(-3adj(2A^{-1}))\right)\right)$ is $2^m.3^n$ .The value of $m+2n =$ ____

A) 36 B) 8 C) 28 D) 44

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70. Line segments are drawn from a fixed point $A(1,4)$, such that their other end lie on a parabola $y^2 = 4x$. If locus of a point (nearer to A) which divide all these line segments in ratio $2:3$, is also a parabola whose focus is at (x_1, y_1) then $x_1 + y_1$ equals_____

A) $\frac{17}{5}$ B) 3 C) $\frac{18}{5}$ D) 4

SECTION – II

(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
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71. Consider the integral $I = \int_0^{10} \frac{[x]e^{[x]}}{e^{x-1}} dx$, where [x] denotes the greatest integer less than or equal to x. Then the value of I is equal to $K(e-1)$, then the value K is equal to_____

72. Let $F: R \to R$ be a thrice differentiable function. Suppose that $F(1) = 0, F(3) = -4 and F(x) < 0$ for all $x \in (1,3)$, $f(x) = xF(x)$ for all $x \in R$.

If $\int_1^3 x^2 F'(x) dx = -12 and \int_1^3 x^3 F''(x) dx = 40$, then $9f'(3) - f'(1) = -K$, then K is equal to?

73. If the area of the region $\{(x,y): |x^2 - 2| \le y \le x\}$ is A and $A = \frac{m}{n}$, m and n are coprime, then m-n is equal to?

74. If N is the number of ways in which 3 distinct numbers can be selected from the set $\{3^1, 3^2, 3^3, ..., 3^{10}\}$ so that they form a G.P. then the value of $N/5$ is

75. If the four distinct points $(6,-4), (22,12), (-2,4)$ and $(13-k, 13+k), k > 0$ lie on a circle of radius r, then the value of $k + r^2$ is_____

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